The form of dissipation shapes what a quantum reservoir remembers

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Dissipation has emerged as a resource for quantum reservoir computing, with its rate as the usual control. Here we show that its form matters as much: it sets how long a record of the past survives, whether new inputs can be written into it, and how strong a signal it leaves to measure. Independent decay shortens memory as loss grows. Collective decay has dark states in which the quantum Zeno effect holds the system, so stronger loss prolongs memory. Whether an input reaches this protected memory depends on the phases with which the qubits couple to the decay channel: at strong loss, a channel that is not aligned with the input recovers the most memory at every measurement budget. For reservoirs that leak slowly and are mixed by their inputs, we prove that at a fixed input strength a stored signal combines on average as many inputs as the disturbance per input times the memory lifetime. A longer memory therefore tolerates only a weaker input, and a weaker input leaves a fainter signal. The form of the dissipation, the strength of the input and the measurement budget thus become joint design parameters of a physical memory.

1 Introduction

Processing information over time requires a memory of what came before. A physical system can provide such a memory because each input changes its state and leaves traces that influence its later response. Reservoir computing uses these traces to perform a task: an input stream drives a dynamical system, and a weighted sum of quantities describing its state produces the output. Only these weights are trained; this mapping is called a linear readout [1, 2]. In quantum reservoir computing (QRC), the evolving system is quantum mechanical, and the readout uses quantities obtained by measuring it [3]. The challenge is to keep earlier inputs accessible while new information continues to arrive.

The environment shapes how these traces fade. A system can lose energy or information to its surroundings through processes collectively described as dissipation [4–15]. Previous work has shown that tuning the dissipation rate improves memory, in some reservoirs with the best performance at an intermediate loss rate [6, 9]. Input, interactions, and dissipation can also be optimized together [16, 17]. The rate specifies how strongly the environment acts, but the way it acts is a separate design choice. Engineering this interaction can prepare and protect selected quantum states [18–21]. For a system processing an ongoing input stream, this raises the possibility of controlling which traces of the past survive.

This distinction becomes concrete for a reservoir of interacting spins, each a two-level quantum system. When the spins decay through separate routes into the environment, they undergo independent decay. When they share a common route, they undergo collective decay. In certain joint states, the different emission paths interfere and cancel, preventing decay through that shared route. These protected states are called dark states [22–24]. Internal dynamics can still move the system into states that do decay. If this decay is sufficiently rapid, however, it suppresses those transitions, a phenomenon known as the dissipative quantum Zeno effect [25–28]. Stronger loss therefore slows the escape of information from the protected states (Fig. 1c).

Survival alone does not make information usable. Each new input changes the same state that

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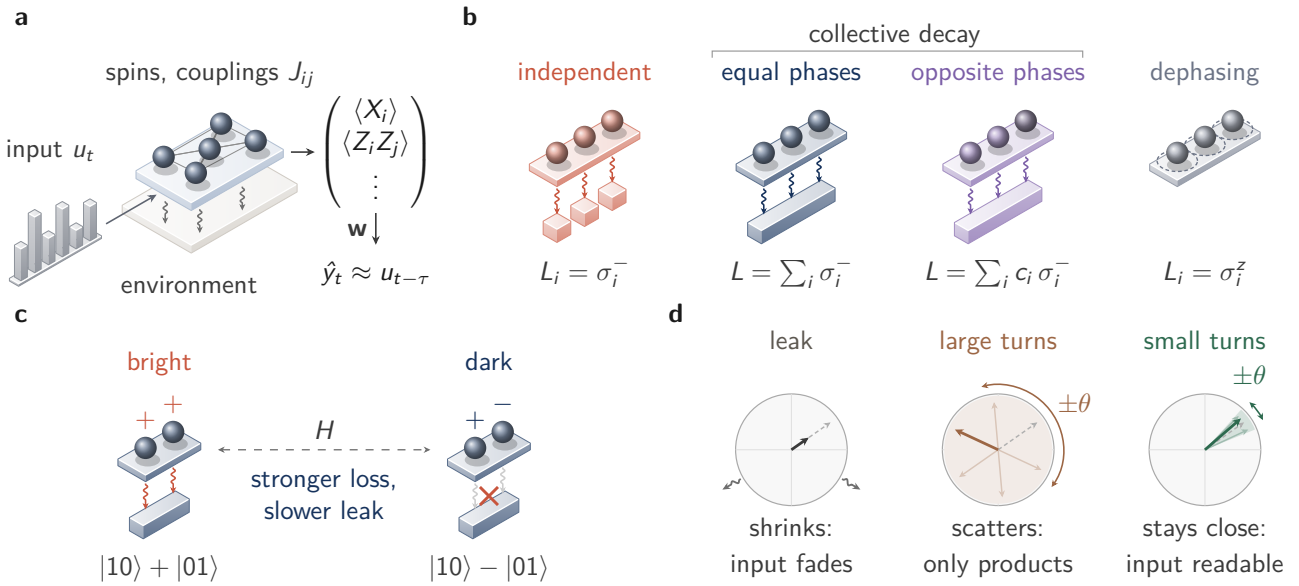


Figure 1: **Engineering how a quantum reservoir leaks. a** Inputs drive interacting spins, and a trained linear readout recalls past inputs. **b** Four forms of dissipation with the same total coupling. The darker spins couple with the opposite sign. **c** Two spins on one channel: their decays add (bright) or cancel (dark), and strong loss holds them in the dark state, the quantum Zeno effect. **d** A record fades because it leaks away or because later inputs turn it at random.

stores the past. A signal that initially records one input therefore comes to depend on combinations of several inputs, such as their product. This mixing is useful for tasks that require relationships between inputs, but it reduces the capacity available for recalling individual inputs [29–31]. Reading the memory has a cost as well: resolving a weak quantum signal requires repeated preparation and measurement [10, 13, 32], so a long-lived record is hard to retrieve if its signal is faint. Previous work has established memory limits in classical reservoirs [2, 33, 34], explored input strength [35], and investigated how noise and protected states affect quantum information processing [36–39]. Closest to our work, Ref. [6] tunes the dissipation rate of the reservoir we study, Ref. [31] finds one trade-off between memory and nonlinearity across reservoir schemes, and Ref. [39] reads the output of a reservoir from a decoherence-free subspace to classify quantum states. The Zeno protection of dark states and the benefit of a weak input for linear memory are established. How protection, continued input, and measurement together determine the part of the past that remains recoverable is open.

Here we show that the form of the dissipation sets three separate properties of a memory: how long a record survives, whether an input can be written into it, and how strong a signal it leaves to measure (Fig. 1). We report four results. First, at equal total coupling the form of the dissipation decides whether stronger loss shortens or prolongs the memory: independent decay shortens it, and collective decay prolongs it through dark states (Sec. 3.1). Second, we prove a law for the mixing of inputs, for reservoirs that leak slowly and are mixed by their inputs: a stored signal combines on average $s\,t_{\mathrm{mem}}$ inputs, the disturbance s caused by each input times the memory lifetime t_{mem} (Sec. 3.2). A longer memory must therefore be written more gently, which we call gentle writing. Third, the phases with which the spins couple to a collective channel decide whether an input reaches the protected memory at all (Sec. 3.3). Fourth, the state that the dissipation leaves behind sets the size of the signal that measurements must resolve: at strong loss a collective channel with non-aligned phases recovers the most memory at every measurement budget, and where the loss can be made weak, independent decay recovers more still (Sec. 3.4).

Together, these results connect the physical survival of information to what a readout can recover from it. The protection relies on quantum interference, while the mixing law holds for every reservoir that meets its conditions, classical ones included, and gives the size of the trade-off between memory and nonlinearity. The form of the dissipation, the strength of the input and the measurement budget thereby become joint

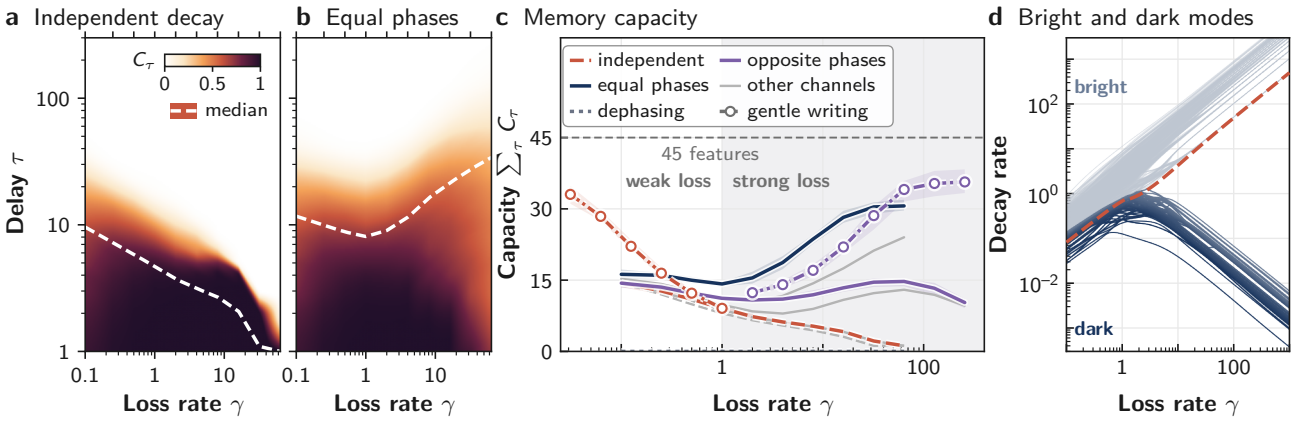


Figure 2: **The decay structure decides whether loss shortens or prolongs the memory. a,b** Recall of the input τ steps back: stronger loss pushes it to shorter delays with independent decay (a) and to longer delays with the equal-phase channel (b). **c** Linear memory capacity of all channels. Shading marks loss stronger than the couplings ($\gamma > 1$). **d** With the equal-phase channel the dark modes decay more slowly as the loss grows, while the slowest mode of independent decay (dashed) decays faster.

design parameters. This makes recoverable memory a quantitative target for engineering physical systems that process information over time.

2 Model

The reservoir is the driven spin system of Ref. [6]. Five spins ($N = 5$) interact through random couplings J_{ij}, and in each step a random input u between 0 and 1 sets the strength of a transverse field for a time Δt:

$$H(u) = \sum_{i<j} J_{ij} X_i X_j + h \sum_i Z_i + h(1+u) \sum_i X_i, \tag{1}$$

with Pauli operators X_i, Z_i of spin i and a field strength h. To weaken the input at a fixed mean drive, we replace $h(1+u)$ by $h[\frac{3}{2} + \eta(u - \frac{1}{2})]$, where the amplitude η scales how far each input moves the field. The state ρ of the spins follows a Lindblad master equation [40, 41],

$$\dot{\rho} = -i[H(u), \rho] + \sum_k \Big(L_k \rho L_k^\dagger - \tfrac{1}{2}\{L_k^\dagger L_k, \rho\} \Big), \tag{2}$$

where the Hamiltonian drives the internal dynamics and each jump operator L_k, scaled by one loss rate γ, is one way the environment changes the state, such as removing an excitation or erasing a phase.

A linear readout of 45 features, the single-spin and two-spin expectation values, is trained to estimate the input from τ steps back. Its recall C_τ, between 0 and 1, is the squared correlation between the estimate and the input on held-out data. The linear memory capacity $\sum_\tau C_\tau$ adds up the recall over all delays. One unit is one past input recalled perfectly, and the ceiling is 45, the number of features. The memory lifetime t_{mem} says how long the variance of a stored signal survives (Appendix A.4). Sections 3.1 to 3.3 use exact expectation values, as if every measurement were repeated infinitely often, and Sec. 3.4 counts measurements. The capacities are numerical results for five spins with all-to-all couplings, while the equal-coupling identity, the slowing of the dark modes, the alignment rule and the mixing law below hold for any number of spins and any Hamiltonian. How the signal scales with the number of spins is open, and reservoirs whose signals concentrate need exponentially many measurements [42].

We compare decay channels at equal total coupling $\mathrm{Tr}\, K = \sum_k \mathrm{Tr}\, L_k^\dagger L_k$ and hold the Hamiltonian fixed to isolate the effect of the dissipative structure. The generator of Eq. (2) has 4^N modes, patterns of change in the state that each decay at their own rate. Equal total coupling fixes their mean rate for every Hamiltonian, but not the slow modes, which keep a record longest and become slower or faster with the channel (Appendix A.2). Table 2 lists the eight channels we studied.

Four of them carry the argument (Fig. 1b). With independent decay every spin loses its excitation on its own. With collective decay all spins lose it through one shared channel, either with equal phases or with opposite phases on spins 4 and 5. We call these the equal-phase and the

opposite-phase channel. With dephasing every spin loses its phase. What tells them apart is whether the dissipation leaves only a single stationary state or a protected subspace that can carry a record. Operators that the dissipation alone leaves unchanged are protected, although the Hamiltonian can still turn them or couple them to decaying ones. Independent decay protects only its ground state. Neither collective channel can empty its dark states, so every operator on them is protected. There are $\binom{N}{\lfloor N/2 \rfloor}$ of them, ten for five spins, and their number grows exponentially with the number of spins. Dephasing keeps the populations, the probabilities of the spin configurations, but erases the coherences between them. Every step then leaves the fully mixed state, with equal populations and no coherence, unchanged whatever the input: the dynamics is unital. The driven reservoir relaxes to this state, after which no input leaves a trace, so dephasing stores nothing.

3 Results

3.1 Dissipative structure controls memory lifetime

Increasing the loss separates the channels into distinct memory regimes (Fig. 2). With independent decay, recall retreats to ever shorter delays and the linear memory capacity falls steadily (Fig. 2a,c). With the equal-phase channel the capacity first dips and then doubles, and at strong loss inputs fifty steps back are still recalled (Fig. 2b,c). The other channels split the same way (Fig. 2c, grey): exchange and gain/loss follow independent decay, two collective channels follow the equal-phase channel, and pairwise decay follows the opposite-phase channel. The decay rates show the split (Fig. 2d). Without protected operators the slowest rate grows in proportion to γ and the memory lifetime shrinks. With dark states the slow rates fall as $1/\gamma$ and the lifetime grows.

Why does stronger collective decay prolong the memory? In the two-spin picture of Fig. 1c, the dark state $|-\rangle \propto |10\rangle - |01\rangle$ can decay only through the bright state $|+\rangle \propto |10\rangle + |01\rangle$, and the Zeno effect [25, 43] suppresses this escape. Its rate falls as Ω^2/γ, where Ω couples $|+\rangle$ and $|-\rangle$, just as the slow mode of an overdamped oscillator settles more slowly the stronger the friction.

This protection follows from the strong-loss

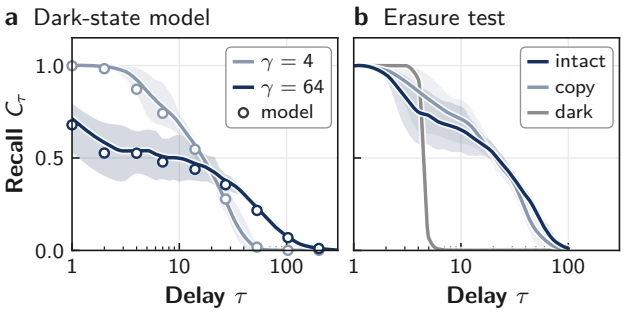


Figure 3: **Dark states hold the memory.** Equal-phase channel. **a** A model of the dark states alone (circles) reproduces the recall of the full reservoir (lines). **b** Erasing what the dark states know about the inputs destroys the memory (dark), while erasing a rotated copy of them leaves most of it (copy).

expansion at fixed system size. At strong loss the reservoir follows its Hamiltonian restricted to the dark states, which turns the stored state but cannot make it decay. Decay enters only at the next order, when the Hamiltonian briefly lifts a dark state into a bright one [26, 28]. One step is then a rotation plus a loss that falls as $1/\gamma$ (Eq. (A2)): a nearly orthogonal map with a small but finite decay rate, the classic recipe for a long memory in a network [33]. Calculations for three to six spins show the same decay out of the dark states, falling as $1/\gamma$ with different prefactors (Appendix E.1). With the equal-phase channel the capacity also grows with the loss at four and six spins (Appendix E.1).

Two tests show that the dark states hold the memory (Appendix E.1). A model of the ten dark states alone, with no fitted parameter, reproduces the recall of the full reservoir at strong loss (Fig. 3a). Replacing the protected components of the state by their time averages after every step, which removes their dependence on the inputs, destroys all memory beyond a few steps, while the same operation on a randomly rotated set of equal size leaves most of it (Fig. 3b). Residual local loss caps this lifetime. For the equal-phase channel, recalling inputs over T steps requires a ratio of collective to local loss rates of at least $4.2\,T^2$ (Appendix E.6): a ratio of 10^2 protects 5 steps and 10^4 protects 49. For superconducting qubits coupled to a waveguide, a four-qubit dark state outlives the decay of a single qubit into the waveguide by more than two orders of magnitude [44], and qubit arrays reach a cooperativity of 172 [45].

Dark states thus protect stored information against decay. With the opposite-phase channel, however, the dark states are just as protected, yet the capacity returns only to its weak-loss level (Fig. 2c). The reason is the ongoing input.

3.2 New inputs mix stored information

Each new input turns the stored state slightly, by an angle that depends on the input (Fig. 1d), so to an old input every newer input acts as noise. Over the memory lifetime the stored direction takes a random walk of small turns $\pm\theta$ and spreads by about $\theta\sqrt{t_{\mathrm{mem}}}$. Once the spread reaches about a radian, the record of an old input survives only multiplied by the inputs that followed it, and a linear readout cannot turn such products back into single inputs.

A stored contribution can therefore depend on one past input or on several jointly. We count the distinct past inputs that it depends on and call this number D. A function of u_{t-3} alone has $D = 1$, even if it is nonlinear in u_{t-3}, and $u_{t-3}u_{t-5}$ has $D = 2$. The information processing capacity [29, 46] also counts how well the readout reconstructs nonlinear functions of past inputs, and sorting its contributions by D separates information about single inputs from information about several jointly (Appendix A.4). The disturbance caused by one input is measured by its mixing fraction s, between 0 and 1: the part of a stored signal that one input turns differently from the average input. For small turns it grows as the square of the turn. The law concerns the covariance of the state, how its components fluctuate together across input histories, and assumes that one pattern of these fluctuations, a single slow direction of the covariance, outlives all others, even when many slow components of the state share it. For reservoirs that leak slowly and are mixed by their inputs, our main theorem then says that a stored contribution involves on average $s\,t_{\mathrm{mem}}$ inputs, the disturbance per input times the memory lifetime:

> **Theorem 1** (Mixing law, made precise in Appendix B)**.** Let a reservoir be reset with probability p per step, so that $t_{\mathrm{mem}} \sim 1/(2p)$ as $p \to 0$, and otherwise turned by an input-dependent rotation. Let the average rotation shrink every state, let the rotations leave no stored covariance unchanged on average except the isotropic one, and let one input change the stationary state (conditions (1) to (3) of Appendix B). Pick a stored contribution with probability equal to its share of the capacity, and let D be the number of distinct inputs it depends on. As $p \to 0$, D/t_{mem} becomes exponentially distributed with mean s, and
>
> $$\mathbb{E}D = s\,t_{\mathrm{mem}} + O(1).$$

The reason is a race: in each step a fraction s of the stored signal picks up one more input, while a fraction $1/t_{\mathrm{mem}}$ of it is lost, so the number of inputs follows a geometric distribution, which becomes exponential for long memories.

Three kinds of evidence support the law. It is proved for the reset model, in any dimension, and nothing in the proof is quantum: classical state-affine reservoirs, whose state is updated by an input-dependent linear map plus an offset [47], obey it under the same conditions. The proof carries over to physical channels without a reset, with the slow decay in place of the reset, when a single direction of the stored covariance is slow (Appendix B.5), while several slow directions keep distinct mixing scales (Appendix B.4). For three spins, interval arithmetic certifies these conditions (Appendix D). For five spins the order recursion, Eq. (A3), gives the distribution of D exactly, and we compare it with the law (Fig. 4c).

At the full input amplitude one input disturbs the stored state only slightly, $s \approx 0.02$. At short memories the mean of D is 1.7 to 1.9, because the fast response of the reservoir combines the latest inputs within a few steps. As the memory grows, the accumulated mixing takes over and the mean of D follows $s\,t_{\mathrm{mem}}$ to within 2.2 inputs, up to 23 inputs at the longest memory. The distribution of D approaches the exponential law at the same time: at $s\,t_{\mathrm{mem}} \approx 20$ it deviates from it by at most 0.07 with the opposite-phase channel and 0.14 with independent decay (Appendix B). Dissipation enters the law through its two factors, the memory lifetime and the disturbance per input, and their product is the size of the trade-off between memory and nonlinearity [31].

The law has a direct consequence. At a fixed input strength, lengthening the memory beyond $s\,t_{\mathrm{mem}} \approx 0.3$ costs recall (Fig. 4a). At the full amplitude the linear memory capacity, summed over

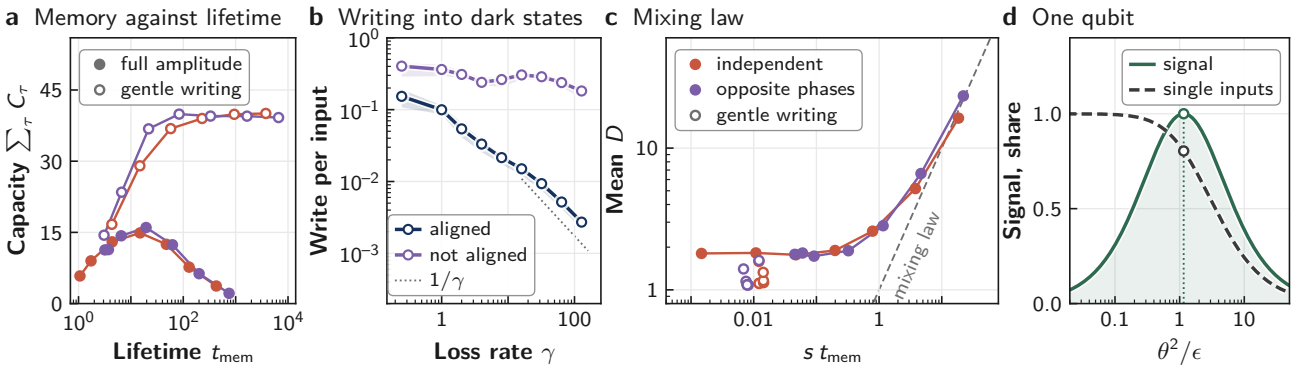


Figure 4: **New inputs mix stored information, and gentle writing limits the mixing. a** At the full input amplitude the capacity, summed over all delays, peaks and then falls as the memory grows. Gentle writing holds it at 40 units. **b** An input aligned with the equal-phase channel writes ever less into the dark states as the loss grows, while a non-aligned input keeps writing. **c** The number of inputs mixed into a stored signal follows the mixing law, and gentle writing keeps it near one. **d** One qubit: the stored signal is largest at an intermediate turn per input (dotted), where single inputs still carry 80% of the capacity.

all delays, peaks at 16 units with the opposite-phase channel, where $s\,t_{\mathrm{mem}} = 0.3$, and falls to 2 units while the lifetime grows from 20 to 750 steps. With independent decay it peaks at 15 units, where $s\,t_{\mathrm{mem}} = 0.2$, and falls to 4. The stored signal is still there, but it has moved into products of inputs: with the opposite-phase channel the share of the capacity carried by single inputs falls from 0.37 to 0.05 (Appendix E.3). Individual inputs remain readable only if the disturbance per input shrinks as the memory lifetime grows. Two controls shrink it: the phases of the decay channel and the amplitude of the input.

3.3 Controlling the disturbance: channel phases and input amplitude

The phases of the collective channel decide whether an input reaches the protected memory at all. Spin i couples to the shared channel with the sign $c_i = \pm 1$, and the dark states are annihilated by the collective lowering operator $\sum_i c_i \sigma_i^-$. We call an input field aligned with the channel if its strength on each spin is proportional to c_i. An aligned field $\sum_i c_i X_i$ is the lowering operator plus its adjoint, so it cannot couple two dark states, for any number of spins and any Hamiltonian. It writes into them only through brief excursions into bright states, with a strength that falls as $1/\gamma$, while the non-aligned part of any other field writes into them directly (Fig. 4b, Appendix E.2). The uniform input of Eq. (1) is aligned with the equal-phase channel, so its disturbance shrinks by itself as the loss grows, and this channel reaches a capacity of 31 units at the full amplitude (Table 3). With the opposite-phase channel the same input is not aligned: it writes directly into the dark states and disturbs them at every loss rate. This is the generic case: only exact alignment keeps the input out of the dark states, and a phase error opens a direct coupling within the dark subspace (Appendix E.2).

The second control is the input amplitude η. That a weak input favours linear memory is known [30, 31, 35]: it removes the products that the fast response forms within a few steps. The mixing law adds what a long memory requires. Because the disturbance s grows as the square of the turn, the accumulated mixing $s\,t_{\mathrm{mem}}$ stays fixed only if the amplitude falls as $1/\sqrt{t_{\mathrm{mem}}}$. We call an input weakened in this way gentle writing. Theorem 1 proves that it is necessary: at a fixed input the share of single inputs vanishes as the memory grows. That it is sufficient is exact in three systems. One qubit has a closed form, three spins have certified bounds (Appendices C and D), and for five spins the recursion shows that single inputs keep between 0.82 and 0.90 of the capacity while the lifetime grows from 20 to 6600 steps (Appendix E.3). The capacity then rises to 40 units and stays there (Fig. 4a).

In the five-spin reservoir the amplitude falls as $1/\sqrt{\gamma}$ for the opposite-phase channel, whose lifetime grows with the loss, and as $\sqrt{\gamma}$ for independent decay, whose lifetime grows as the loss weakens (Fig. 2c, circles). This holds $s\,t_{\mathrm{mem}}$ at 0.01. Within the 300 delays that Fig. 2c sums, the linear memory capacity rises from about 15 to 35

units for both channels (Table 3). Both effects of a weak input contribute: with the opposite-phase channel the accumulated mixing accounts for 43% of the inputs that gentle writing removes from the mean of D at a lifetime of 20 steps, for 84% at 200 steps and for 99% at 750. Since the capacities of all orders add up to the 45 features [29], the gain moves capacity from combinations of inputs to individual inputs. Gentle writing therefore serves recall. A task that needs products of inputs calls for a larger disturbance, and $s\,t_{\mathrm{mem}}$ says how many inputs a stored signal then combines. A long memory therefore needs no dark states: weak independent decay with gentle writing reaches one too. What dark states add is a memory that stays long under strong dissipation and grows with the loss.

This amplitude is a ceiling and not an optimum. The ceiling limits the number of inputs mixed, not the capacity of an exact readout: on a map of loss rates and amplitudes the capacity keeps rising as the input weakens, up to 43 of the 45 units (Appendix E.4). A weaker input mixes less, and without measurement noise a fainter record costs nothing. The lifetime thus sets only a ceiling on the input strength, $s\,t_{\mathrm{mem}} \lesssim 1$. One qubit shows what a weak input gives up (Fig. 4d, Appendix C). Its stored signal is largest when the squared turn per input, accumulated over one relaxation time, is of order one, and it vanishes in proportion to this number for weaker inputs.

For three spins with the opposite-phase channel, computer-checked interval bounds confirm both regimes at strong loss: at a fixed amplitude single inputs lose their share of the capacity, and with an amplitude $\eta_0/\sqrt{\gamma}$ they keep it (Appendix D). The record is written into coherences between the energy levels of the Hamiltonian restricted to the dark states. Coherence shapes the performance of spin reservoirs [48], and erasing these coherences after every input collapses the stored signal as $1/\gamma^2$. A gentler input, however, writes a fainter record. With exact readout this costs nothing. With measurements it does.

3.4 Measurement limits recoverable memory

Exact expectation values remove the cost of resolving weak signals. A memory that is written more gently grows longer, not larger: it cannot exceed the 45 features, so each delay is recalled more faintly. With projective readout each feature is estimated from n preparations of the reservoir per measurement setting and step, and its statistical error must stay below the signal, the variation of the feature with the inputs (Appendix A.5). We call the capacity that survives this sampling recoverable. It is the counterpart, for the recall of past inputs, of the resolvable expressive capacity of Ref. [32].

The size of the signal is set by the state that the dissipation leaves behind. An input acts on the stationary state ρ_* through the commutator with its Hamiltonian, and only the part of this change that lies on the slow modes is remembered. We call its size the write strength. It is a linear response of the stationary state to the input field, in the spirit of the link between memory and optical absorption of Ref. [9]. When all channels are compared at one decay-only lifetime, the feature variance of the seven channels that write follows the write strength over four orders of magnitude (Fig. 5b). So does the fraction of the exact capacity that 10^6 preparations per setting and step recover (Fig. 5c): over all 56 reservoirs the rank correlation between the two is 0.96. Collective decay with equal phases leaves a polarized state that the uniform input moves only through bright states. It has the largest exact capacity of all channels, 35 units, and recovers 1% of it. Reversing the coupling phase of two spins lets the input move the state directly (Sec. 3.3), and the opposite-phase channel recovers 60% of its 25 units. Independent decay recovers 73% of 24 (Appendix E.5). The exact capacity therefore does not tell what measurements recover. The write strength tells which fraction of it they do.

The form matters most where the loss is strong (Fig. 5a). At $\gamma = 64$ and the full amplitude, 10^4 preparations per setting and step recover 8 units with the opposite-phase channel and 0.2 units, the floor of no recall, with the equal-phase channel and with independent decay. With 10^6 preparations the three recover 10, 2.3 and 0.2 units (Table 3). A collective channel with non-aligned phases is therefore the right choice when strong loss is given, as for emitters that radiate into a common mode [22–24]. When the loss can be made two thousand times weaker, independent decay recovers more: 11 units from 10^4 preparations at $\gamma = 1/32$.

A short memory needs few preparations. With 10^3 per setting and step, the opposite-phase chan-

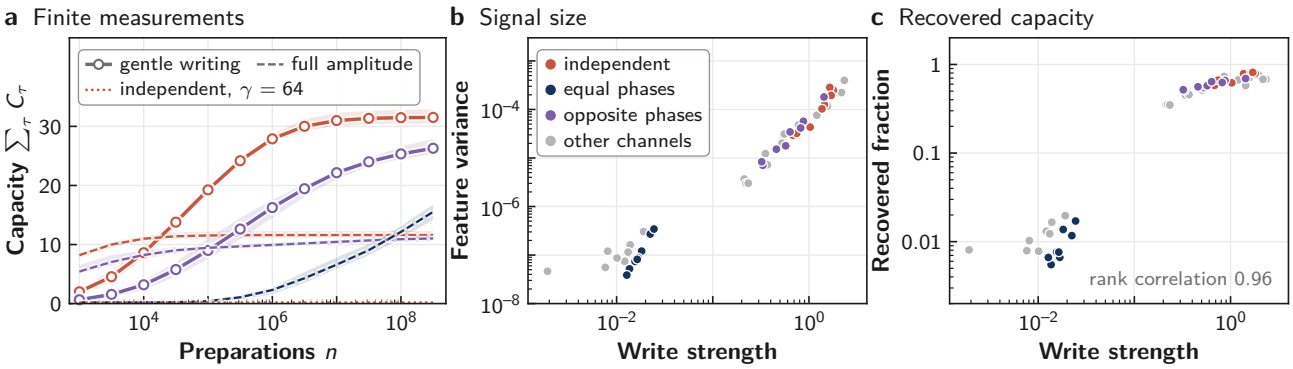


Figure 5: **Measurement limits recoverable memory.** **a** Capacity recovered against the number of preparations per setting and step, with replayed inputs: independent decay at weak loss ($\gamma = 1/32$) and the two collective channels at strong loss ($\gamma = 64$). **b** The signal grows with the write strength of the stationary state. **c** So does the fraction of the exact capacity that 10^6 preparations recover. Points in b and c are single reservoirs at a matched decay-only lifetime, with the full input history.

nel at strong loss recovers 5 units and weak independent decay 8. Gentle writing pays at larger numbers: its capacity overtakes that of the full amplitude at 2×10^4 preparations with independent decay and at 10^5 with the opposite-phase channel, and reaches 31 and 26 units at 3×10^8.

Every preparation replays the recent inputs before it is measured [13], and the replayed window grows with the memory lifetime. When every replayed step is counted (Appendix A.5), a budget of 10^9 reservoir steps per output step gives the three forms of dissipation the same memory at the full amplitude: 13 units with independent decay at $\gamma = 1/8$, 13 with equal phases at $\gamma = 1$ and 12 with opposite phases at $\gamma = 16$ (Appendix E.5). A non-aligned collective channel thus holds at a hundredfold stronger loss what independent decay holds at weak loss. With gentle writing, weak independent decay recovers the most, 34 units against 21.

With measurements a best amplitude exists. With 10^6 to 10^8 preparations per setting and step it falls with the decay-only lifetime, with fitted exponents between -0.40 and -0.46, close to the $-1/2$ of the ceiling (Appendix E.4). The lifetime sets the ceiling on the input strength, and the measurement budget sets the floor.

4 Conclusion

A useful quantum reservoir must retain a readable record of the past while continuing to accept new information. We have shown that the structure of dissipation changes how this record survives. Independent decay shortens memory as loss grows, whereas strong collective decay prolongs it by protecting information in dark states through the quantum Zeno effect. The environment therefore provides a physical means of designing the memory of a driven quantum system.

Longer storage also leaves more time for subsequent inputs to mix with the stored information. Our mixing law makes this accumulation quantitative. It sets a ceiling on the input strength: the longer the memory, the more gently new information must be written, so that earlier inputs remain individually recoverable. In the spin reservoirs studied, weakening the input raises the capacity to recall past inputs from about 15 to 35 units with exact readout, where no input is too weak.

Measurement adds a further requirement: the stored information must produce a signal strong enough to resolve. Protection does not make a memory readable: the equal-phase channel holds the largest memory and leaves the faintest signal. At strong loss a collective channel with non-aligned phases keeps a readable memory where independent decay keeps none. Where the loss can be made weak, independent decay recovers the most because its signals are stronger. The resources needed to retrieve information are therefore part of the design of the memory itself.

Engineering the environment, the input, and the readout together thus provides a route from long-lived quantum dynamics to memory that a readout can recover. Four rules summarize this design:

(1) Collective decay protects a memory in dark states. Stronger loss then makes it last longer, as long as local loss stays weak.

(2) The input must be able to reach the protected memory. An input aligned with the decay channel writes into it ever more weakly as the loss grows.

(3) The longer a memory lasts, the more gently each input must be written. Otherwise new inputs mix with the old ones.

(4) What can be read out depends on how strongly the inputs move the state that the dissipation leaves behind. At strong loss a collective channel that is not aligned with the input recovers the most.

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Data availability

The code, the data behind every figure, and the derivation and certificates of the three-spin limits are at https://github.com/eybmits/what-a-dissipative-quantum-system-remembers.

Author contributions

M.B. contributed to conception, methodology, software, analysis, visualization and writing, I.F. and J.W. to methodology, interpretation and editing, and C.L.-P. and J.S. to supervision and conceptual development. OpenAI Codex and Claude Code (Anthropic) assisted with manuscript preparation, checks of derivations and references, and simulation, verification and plotting code. The authors reviewed the outputs and take responsibility for the final content.

A Methods

This appendix specifies the five-spin simulations and defines the quantities that the main text uses.

Table 1: Quantities and terms of the text. Times are in input steps of length Δt.

t_{mem}	memory lifetime: how long the variance of a stored signal survives, whatever inputs it combines (Appendix A.4)
t_{f}	single-input recall time: how long one input stays linearly readable on its own (Appendix A.3)
t_{leak}	decay-only lifetime: the recall time with the input held at its mean value (Appendix A.5)
τ_{90}	replayed range: the first τ_{90} delays hold 90% of the exact recall
D	number of distinct past inputs that a stored contribution depends on
s	disturbance per input: the part of a stored signal that one input turns differently from the average input
aligned	an input field whose strength on each spin follows the phases of the collective channel
gentle writing	an input amplitude that falls as $1/\sqrt{t_{\mathrm{mem}}}$
write strength	how strongly an input changes the slow modes of the stationary state
recoverable	the capacity that survives sampling with n preparations per setting and step

Table 1 lists the quantities and terms of the text.

A.1 Reservoir, readout and capacity

The Hamiltonian of Eq. (1) uses $h = 1/2$ and couplings J_{ij} drawn uniformly from $[-1, 1]$, so the transverse field runs from h to $2h$. Each input u is drawn uniformly from $[0, 1]$ and held for a step of length $\Delta t = 1/2$. We integrate Eq. (2) in the 1024-dimensional basis of Pauli operators and evaluate the step map $E(u) = e^{\Delta t \mathcal{L}(u)}$ by Chebyshev interpolation in u at ten nodes. Each reservoir draws its couplings and its input stream from one seed, and every channel uses the same eight seeds. After a washout of $\max(800, 60\gamma)$ steps, at least four times the single-input recall time, we record the 45 features, the expectation values of X_i, Y_i, Z_i and of X_iX_j, Y_iY_j, Z_iZ_j, for 2000 training, 400 validation and 1600 test inputs. The readout is a least-squares fit with an intercept on the training and validation inputs, and C_τ is the squared correlation of its estimate with $u_{t-\tau}$ on the test inputs. The linear memory capacity sums C_τ over delays 1 to 300. A delay without any recall reads about $1/1600$ on the test inputs, so the sum has a floor of about 0.2 units. Figures show the mean of the eight reservoirs, with bands from the 16th to the 84th percentile.

Table 2: Decay channels at five spins, each with $\operatorname{Tr} K = 80\gamma$, and the number of operators that the dissipator leaves unchanged. The operators $\sigma_i^{\mp}$ lower and raise spin i, with $i < j$ for pairwise decay, $i \neq j$ for exchange and $c = (1, 1, 1, -1, -1)$.

channel	jump operators	protected operators
independent	$\sqrt{\gamma}\,\sigma_i^-$	1
equal phases	$\sqrt{\gamma}\sum_i \sigma_i^-$	100
opposite phases	$\sqrt{\gamma}\sum_i c_i\sigma_i^-$	100
two channels	$\sqrt{\gamma}\sum_{i\leq 3}\sigma_i^-$, $\sqrt{\gamma}\sum_{i>3}\sigma_i^-$	36
pairwise	$\sqrt{\gamma}\,\sigma_i^-\sigma_j^-$	36
exchange	$\sqrt{0.4\gamma}\,\sigma_i^-$, $\sqrt{0.3\gamma}\,\sigma_i^-\sigma_j^+$	1
gain/loss	$\sqrt{\gamma/1.3}\,\sigma_i^-$, $\sqrt{0.3\gamma/1.3}\,\sigma_i^+$	1
dephasing	$\sqrt{\gamma/2}\,Z_i$	32

A.2 Decay channels and equal coupling

Table 2 lists the jump operators of the eight decay channels. Their weights give every channel the same total coupling $\operatorname{Tr} K = N2^{N-1}\gamma = 80\gamma$, which fixes the mean decay rate. For a traceless jump L the superoperator trace of its dissipator is $-2^N \operatorname{Tr} L^\dagger L$, while the commutator with any Hamiltonian is traceless, so the 4^N eigenvalues λ of the generator obey

$$\frac{1}{4^N}\sum \operatorname{Re}\lambda = -\frac{\operatorname{Tr} K}{2^N} \tag{A1}$$

for every channel and every Hamiltonian. The protected operators form the kernel of the dissipator. Their number is not the number of dark states: the ten dark states of a collective channel carry $10 \times 10 = 100$ protected operators.

A.3 Slow modes, single-input recall time and the strong-loss generator

The input-averaged step $\bar{E} = \mathbb{E}_u E(u)$ is computed by six-point Gauss–Legendre quadrature. Its eigenvalues λ_j give decay rates $\kappa_j = -\ln|\lambda_j|/\Delta t$. If the dissipator protects n operators, the slow modes are the $n-1$ slowest decaying ones besides the stationary state, and the single-input recall time is $t_{\mathrm{f}} = 1/(\Delta t\,\bar{\kappa})$, with $\bar{\kappa}$ their mean rate. Without protected operators we use the slowest mode. The averaged step carries the part of a stored signal that still depends on a single input. Its mean rate adds decay and mixing to second order in the amplitude, $\bar{\kappa} = \bar{\kappa}_0 + c\,\eta^2 + O(\eta^4)$, because the inputs are distributed symmetrically about their mean and the term of $\bar{E}$ linear in η vanishes. Appendix E.2 tests this sum numerically.

Figure 2d shows instead the decay rates $-\operatorname{Re}\lambda$ of the generator at the mean input, $u = 1/2$. With a collective channel we call the $n-1$ slowest of them dark and all faster ones bright.

Let $\mathcal{D}$ be the unit-rate dissipator, $\mathcal{P} = \lim_{t\to\infty} e^{t\mathcal{D}}$ the projector onto its kernel and $\mathcal{S} = (\mathcal{D}+\mathcal{P})^{-1} - \mathcal{P}$ its inverse on the decaying operators. At strong loss, eliminating the decaying operators to second order [26, 28] gives the generator of the protected coordinates,

$$\mathcal{G}(u) = \mathcal{P}\mathcal{H}(u)\mathcal{P} - \frac{1}{\gamma}\,\mathcal{P}\mathcal{H}(u)\mathcal{S}\mathcal{H}(u)\mathcal{P}, \tag{A2}$$

with $\mathcal{H} = -i[H, \cdot\,]$. The first term is the Hamiltonian restricted to the protected operators, which turns them, and the second is the loss through brief excursions into bright states.

Gentle writing uses the amplitude $\eta = \min(1, \gamma^{-1/2})$ of Sec. 2 for the opposite-phase channel and $\eta = \min(1, \gamma^{1/2})$ for independent decay.

A.4 Number of inputs mixed

The order D of a stored signal is the number of distinct past inputs it depends on (Sec. 3.2). Precisely, for independent inputs, every function of the past inputs with finite variance splits uniquely into orthogonal parts, each of which depends on a fixed finite set of inputs. The order of a part is the size of its set, not a polynomial degree. For signs $u_t = \pm 1$ the parts are the products $\prod_{j\in J} u_{t-j}$, and for continuous inputs one uses orthogonal polynomials in each input. We collect the traceless coefficients of the state in an orthonormal operator basis in a vector r, on which quantum channels act affinely [49, 50], $r_t = A(u_t)r_{t-1} + b(u_t)$. All means and covariances are taken over input histories. Let $\bar{A} = \mathbb{E}A$ be the mean step, m the stationary mean and $\xi = A(u)m + b(u) - m$ the innovation, the change that one input writes, with covariance V. With $\mathcal{B}X = \bar{A}X\bar{A}^{\mathsf{T}}$, $\mathcal{T}X = \mathbb{E}[AXA^{\mathsf{T}}]$ and $\mathcal{K} = \mathcal{T} - \mathcal{B}$, the covariance G_k carried by products of order k obeys

$$G_1 = (1-\mathcal{B})^{-1}V, \qquad G_{k+1} = (1-\mathcal{B})^{-1}\mathcal{K}G_k. \tag{A3}$$

The innovation adds the current input alone. The mean step carries a stored product along without adding an input. The fluctuating part $\mathcal{K}$ of the step multiplies it by the current input, which raises its order by one. The G_k add up to the

stationary covariance Σ. A readout of features $y = Mr$ has the capacity $\mathrm{Tr}[(M\Sigma M^\mathsf{T})^+ M G_k M^\mathsf{T}]$ of order k, and these capacities add up to the rank of the readout [29]. Divided by their sum, they give the share of each order, the distribution of D in Theorem 1. For one observable with coordinates a the share of order k is $a^\mathsf{T} G_k a / a^\mathsf{T}\Sigma a$, and the share of single past inputs is $a^\mathsf{T}\bar{A}G_1\bar{A}^\mathsf{T}a/a^\mathsf{T}\Sigma a$.

The memory lifetime and the mixing fraction come from the slowest mode of the covariance step $\mathcal{T}$: its eigenvalue is $1 - 1/t_{\mathrm{mem}}$, and with its right eigenvector Q and left eigenvector ℓ the mixing fraction is $s = \ell(\mathcal{K}Q)/\ell(Q)$, as in Appendix B.5. For five spins we evaluate Eq. (A3) on the 1023 traceless Pauli coordinates (Appendix E.3).

A.5 Finite sampling and the measurement budget

A quantum measurement gives a random outcome, so every feature is an average over n preparations per measurement setting and step. We use three settings, with all spins read along X, Y or Z. For each step we form the outcome distributions of the three settings and draw n preparations from each. The readout is fitted as above, with a ridge chosen from fourteen values between 0 and 10^4 on the validation inputs. In Sec. 3.4 the distributions come from the replay: each output step replays its last τ_{90} inputs from $|00000\rangle$, where the first τ_{90} delays hold 90% of the exact recall. We count the budget B in reservoir steps, preparations times replayed steps summed over the three settings, per output step, so that $n = B/(3\tau_{90})$, and we also run $n = 10^3$ to $10^{8.5}$ in half decades. Only Fig. 5b,c and Table 4 use the state with the full history. Its matched decay-only lifetime is the leak time $t_{\mathrm{leak}} = 1/(\Delta t\,\bar{\kappa}_0)$ of 32 input steps, where $\bar{\kappa}_0$ is the mean decay rate of the slow modes of the generator at the mean input, without the mixing by the inputs (Appendix E.5).

B The mixing law

This appendix proves the mixing law of Sec. 3.2 and extends it to physical channels without a reset, which Appendix D needs.

B.1 Setting and statement

We describe the reservoir by the n traceless coordinates r of its state in an orthonormal basis of operators, the Bloch vector for a qubit. A unitary acts on r as a rotation. In each step the reservoir is reset to a fixed state with a small probability p and is otherwise rotated by $O(u)$:

$$r_t = (1-p)\,O(u_t)\,r_{t-1} + p\,b. \tag{B1}$$

Here b represents the reset state. Equation (B1) averages over whether the reset occurred, so the only remaining randomness is the input history, with independent and identically distributed inputs u_t. A stored signal loses a fraction $1-(1-p)^2 = 2p-p^2$ of its variance per step, so its memory lifetime, defined by the slow eigenvalue $1 - 1/t_{\mathrm{mem}}$ of the covariance step (Appendix A.4), is $t_{\mathrm{mem}} = 1/(2p-p^2)$, or $1/(2p)$ to leading order. Three conditions make the law hold:

(1) *The average step forgets.* The average rotation $A = \mathbb{E}O$ shrinks every vector: its spectral radius is below one.

(2) *The inputs mix.* The only covariance matrices that the random rotations leave unchanged on average, $\mathbb{E}[OXO^\mathsf{T}] = X$, are multiples of the identity.

(3) *The input writes.* One input changes the state: $\mathbb{E}\|(O-A)\mu\|^2 > 0$ with $\mu = (1-A)^{-1}b$.

The first condition means that only products of inputs can be stored for long, the second that a stored signal spreads evenly over all directions, and the third that there is something to store. The mixing fraction

$$s = 1 - \frac{\|A\|_F^2}{n} \tag{B2}$$

says how strongly one input turns the stored state. For a unit vector x of random orientation, $\mathbb{E}\|(O-A)x\|^2 = s$: one input turns a fraction s of a stored signal away from the average rotation. The fraction lies between 0 and 1, and the first condition makes it positive. If the inputs $u = \pm 1$ rotate by $O = e^{u\theta G}$ with a fixed generator G, whose eigenvalues are $i\lambda_1, \ldots, i\lambda_n$, the average rotation is $\cosh\theta G$ and $s = \frac{1}{n}\sum_j \sin^2\theta\lambda_j$, which grows as θ^2 for small turns.

Mixing law, precise form. Assume conditions (1)–(3) and take any fixed linear readout. Pick a stored contribution with probability equal to its share of the readout's capacity, and let D be its order, the number of distinct inputs it depends on. As $p \to 0$, the ratio D/t_{mem} becomes exponentially distributed with mean s, and

$$\mathbb{E}D = s\, t_{\mathrm{mem}} + O(1).$$

Equivalently, products of more than $x\, t_{\mathrm{mem}}$ inputs carry the share $e^{-x/s}$ of the capacity.

B.2 The idea: a race

In the race of Sec. 3.2, on average $1/s$ steps pass before the next input joins a product, and meanwhile it loses a fraction $1/(s\, t_{\mathrm{mem}})$ of its weight, so products of more than K inputs carry about

$$\Big(1 - \frac{1}{s\, t_{\mathrm{mem}}}\Big)^{K} \approx e^{-K/(s\, t_{\mathrm{mem}})},$$

an exponential law with mean $s\, t_{\mathrm{mem}}$. The proof makes this bookkeeping exact.

B.3 Proof

The proof works with covariance matrices. Appendix A.4 splits the stationary covariance Σ of the state into the parts G_k carried by products of k inputs and gives their recursion, Eq. (A3): the mean step $\mathcal{B}$ carries a stored product along, and the fluctuating part $\mathcal{K}$ of the step adds one input to it. In the reset model the mean step is $\mathcal{B}X = (1-p)^2 AXA^{\mathsf{T}}$, the full step $\mathcal{T}X = (1-p)^2\, \mathbb{E}[OXO^{\mathsf{T}}]$, and $\mathcal{K} = \mathcal{T} - \mathcal{B}$. We write $\mathcal{B}_0$, $\mathcal{T}_0$ and $\mathcal{K}_0$ for the same maps at $p = 0$.

Step 1: the stored signal is small and spread evenly. The stationary mean is $m = p\,[1 - (1-p)A]^{-1} b = p\mu + O(p^2)$, so one input writes the change $(1-p)(O-A)m$, of size p. Its covariance V has the trace $p^2 v + O(p^3)$ with $v = \mathbb{E}\|(O-A)\mu\|^2$, which is positive by condition (3). The random rotations preserve the trace of a covariance and, by condition (2), leave only the identity unchanged. Since they cannot enlarge a covariance, the identity is a simple fixed point with the projector $X \mapsto (\operatorname{Tr} X/n)\, 1$. On every other direction $(1-\mathcal{T})^{-1}$ stays bounded as $p \to 0$, while on the identity it equals $1/[1-(1-p)^2] \approx 1/(2p)$. The written covariance therefore piles up along the identity:

$$\Sigma = (1-\mathcal{T})^{-1} V = \frac{pv}{2n}\, 1 + O(p^2). \tag{B3}$$

Step 2: few inputs carry a vanishing share. The mean step forgets by condition (1), so $(1-\mathcal{B})^{-1}$ stays bounded as $p \to 0$. The recursion then makes $G_1 = (1-\mathcal{B})^{-1} V$ of order p^2, and the same holds for every G_k with fixed k. Measured against Σ, which is of order p and has full rank, any fixed set of orders carries a share of order p of the capacity of any readout. A long memory consists of products of many inputs.

Step 3: counting the inputs. Call $\mathcal{F} = (1-\mathcal{B})^{-1}\mathcal{K}$ the order transfer, so that $G_{k+1} = \mathcal{F} G_k$. At $p = 0$ it leaves the identity unchanged, because $\mathcal{K}_0 1 = 1 - AA^{\mathsf{T}} = (1 - \mathcal{B}_0)1$. Its matching left eigenvector is the weight

$$\ell(X) = \operatorname{Tr}[(1 - A^{\mathsf{T}}A)X], \qquad \ell(1) = ns, \tag{B4}$$

since $\ell(\mathcal{F}_0 X) = \operatorname{Tr} \mathcal{K}_0 X = \ell(X)$ when the rotations preserve the trace. For small p the eigenvalue one moves to

$$1 - \frac{2}{s}\, p + O(p^2)\,: \tag{B5}$$

each step up in order costs a fraction $2p/s = 1/(s\, t_{\mathrm{mem}})$ of the weight, as in the race. To see this, differentiate $(1-\mathcal{B})\mathcal{F} = \mathcal{K}$ at $p = 0$ and apply it to the identity, which gives $(1-\mathcal{B}_0)\mathcal{F}_0' 1 = -2 \cdot 1$. Its trace is the weight $\ell(\mathcal{F}_0' 1) = -2n$, and dividing by $\ell(1) = ns$ gives the rate $2/s$.

To turn the eigenvalue into a distribution, weigh order k with $e^{-\lambda p k}$. Summing the recursion gives $\sum_k z^k G_k = z(1 - z\mathcal{F})^{-1} G_1$, and as $z = e^{-\lambda p}$ approaches one, only the eigenvalue of Eq. (B5) makes the inverse large. With $\ell(G_1) = p^2 v + O(p^3)$ and $\beta = 2/s$,

$$\sum_{k \geq 1} e^{-\lambda p k} G_k = \frac{pv}{2n}\, \frac{\beta}{\lambda + \beta}\, 1 + O(p^2). \tag{B6}$$

Divided by Eq. (B3), this is $\beta/(\lambda+\beta)$, the Laplace transform of an exponential distribution of pD with rate β. So $D/t_{\mathrm{mem}} = 2pD\,[1 + O(p)]$ is exponential with mean s in the limit. Because both covariances are multiples of the identity to leading order, the same holds for the share of the capacity of any readout. Finally, $\sum_k k G_k =$

$(1-\mathcal{F})^{-1}\Sigma = \Sigma/(\beta p) + O(p)$ gives $\mathbb{E}D = s\,t_{\mathrm{mem}} + O(1)$.

Two technical points. On symmetric matrices, where covariances live, the eigenvalue one of $\mathcal{F}_0$ is simple: a fixed point satisfies $\mathcal{T}_0 X = X$ and is therefore a multiple of the identity, and $\ell(1) > 0$ rules out a Jordan chain. For $p > 0$ every eigenvalue of $\mathcal{F}$ lies inside the unit circle, by a Perron–Frobenius argument: $\mathcal{F}$ maps positive matrices to positive matrices, and so does $(1-\mathcal{F})^{-1} = 1 + (1-\mathcal{T})^{-1}\mathcal{K}$. The eigenvalues of $\mathcal{F}_0$ therefore lie in the closed unit disk, only the simple one equals one, and only the eigenvalue of Eq. (B5) makes the sum diverge.

B.4 When the law needs one slow direction

Condition (2) matters. Let a classical flag, set by the reset state and never changed, select one of two qubit rotations about Z, by $\pm\pi/6$ or by $\pm\pi/3$, and let the reset state put the flag in either value with probability $\frac{1}{2}$ and the qubit along $+X$ in both. Each sector obeys the law with its own mixing fraction, $1/4$ and $3/4$, and a single Y readout sees their mixture, weighted by the variance each sector stores. The gentler sector stores more, so as $p \to 0$ products of more than $x\,t_{\mathrm{mem}}$ inputs carry the share $w\,e^{-4x} + (1-w)\,e^{-4x/3}$ with $w = 0.82$, which exact moments reproduce. One observable then shows two order scales.

B.5 Physical channels without a reset

A physical channel does not reset to a prescribed state, but the proof used the reset only through three facts: one covariance direction is slow, it leaks at a small rate, and the mean step forgets everything else. The same three steps therefore prove the law for general channels.

Setting. One step is $r_t = A_u(\epsilon) r_{t-1} + b_u(\epsilon)$ on the coordinates that the inputs can reach, with a small leak ϵ, and $\mathcal{B}$, $\mathcal{T}$ and $\mathcal{K}$ are formed from A_u as before. We assume:

(i) A_u and b_u depend smoothly on ϵ in mean square, and b_u vanishes at $\epsilon = 0$.

(ii) The mean step forgets at $\epsilon = 0$: $\mathbb{E}A_u(0)$ has spectral radius below one.

(iii) The channel is mean-square stable for small $\epsilon > 0$: the covariance step $\mathcal{T}$ has spectral radius below one.

(iv) One covariance direction is slow: $\mathcal{T}$ has a simple eigenvalue $1 - \kappa\epsilon + O(\epsilon^2)$ with $\kappa > 0$ and no other eigenvalue near one. At $\epsilon = 0$ its right eigenvector is a positive matrix Q, and its left eigenvector is a weight ℓ that is nonnegative on positive matrices, with $\ell(Q) = 1$.

(v) The input writes along it: the change written by one input, of order ϵ, has covariance $\epsilon^2 V_* + O(\epsilon^3)$ with $\ell(V_*) > 0$.

The reset model is the case $\epsilon = p$, $\kappa = 2$, $Q = 1$ and $\ell = \mathrm{Tr}\,/n$. Condition (iv) concerns the covariance that the random steps carry, not the number of slow modes: at a fixed amplitude the inputs turn the 99 slow dark modes of the opposite-phase channel differently in each step and spread a stored signal over all of them, so that one covariance direction stays slow. Conditions (ii) and (iii) say that the reservoir forgets its starting state on average and in mean square, (iv) that the random steps spread a stored signal over the slow modes faster than it leaks, and (v) that the surviving direction is written.

Mixing law without a reset. Assume (i)–(v) and take a fixed readout $y = Mr$ whose projected slow covariance MQM^{T} has full rank. Then the mixing fraction $s = \ell(\mathcal{K}_0 Q)$ is positive and at most one, and ϵD becomes exponentially distributed with rate κ/s. The mean order is $s/(\kappa\epsilon) + O(1)$: the mixing fraction times the memory lifetime $1/(\kappa\epsilon)$.

Proof. Steps 1 and 2 carry over. The stationary mean is of order ϵ by (i) and (ii), and the slow eigenvalue piles the written covariance up along Q,

$$\Sigma = \epsilon\,\frac{\ell(V_*)}{\kappa}\,Q + O(\epsilon^2),$$

and every fixed order carries a covariance of order ϵ^2. Here the readout condition enters: the shares of the orders in the capacity of the readout follow from Σ only if MQM^{T} has full rank. For one observable a this requires $a^{\mathsf{T}} Q a > 0$. In step 3, $\mathcal{F}_0$ has the same fixed points as $\mathcal{T}_0$, so Q spans them, and $\ell \circ (1 - \mathcal{B}_0)$ is its left eigenvector, with value $\ell(\mathcal{K}_0 Q) = s$ on Q. Differentiating $(1-\mathcal{B})\mathcal{F} = \mathcal{K}$ at $\epsilon = 0$ and applying it to Q gives $(1-\mathcal{B}_0)\mathcal{F}_0' Q = \mathcal{T}_0' Q$, whose weight $\ell(\mathcal{T}_0' Q) = -\kappa$ is the rate of the slow eigenvalue of $\mathcal{T}$. The order eigenvalue is therefore $1 - (\kappa/s)\epsilon + O(\epsilon^2)$, and the Laplace

transform follows as before. It remains to show that s is positive, which also makes the eigenvalue simple. Because $\mathcal{K}_0$ maps positive matrices to positive matrices and ℓ is nonnegative on them, $\ell(\mathcal{B}_0 Y) \le \ell(\mathcal{T}_0 Y) = \ell(Y)$ for every positive Y, so $0 \le \ell(\mathcal{B}_0^j \mathcal{K}_0 Q) \le s \le 1$ for every j. If s vanished, the series $\mathcal{F}_0 = \sum_j \mathcal{B}_0^j \mathcal{K}_0$ would give $\ell(\mathcal{F}_0 Q) = 0$, which contradicts $\mathcal{F}_0 Q = Q$ and $\ell(Q) = 1$.

The spin reservoirs. For three spins the certificates of Appendix D check these conditions. For five spins we check the law itself (Sec. 3.2 and below).

B.6 Numerical check

We compared the law with exact moments for 24 reset reservoirs and readouts in dimensions two to four: two families of three random unitaries, two reset states (a basis state and a random pure state) and two readouts (all coordinates or one random direction). The largest deviation from the law shrinks in proportion to p, from 0.08 at $p = 3\%$ to 0.003 at $p = 0.1\%$. For five spins the share of the capacity in products of more than K inputs approaches $e^{-K/(s\, t_{\mathrm{mem}})}$ as the memory grows: its largest deviation, over K and over the reservoirs, falls from 0.33 and 0.43 at $s\, t_{\mathrm{mem}} \approx 1$ to 0.07 and 0.14 at $s\, t_{\mathrm{mem}} \approx 20$, with the opposite-phase channel and with independent decay.

C One qubit in closed form

This appendix solves the qubit of Sec. 3.3 exactly.

The model. The qubit relaxes towards $+X$ at a rate ϵ, through the jump $\sqrt{\epsilon}\,|+x\rangle\langle -x|$, and the input $u = \pm 1$ turns it about Z by $u\theta$ per unit time. Each input lasts one unit of time, and the qubit is read along Y. In the Bloch plane,

$$\dot{x} = -\epsilon(x-1) - u\theta y, \qquad \dot{y} = u\theta x - \tfrac{1}{2}\epsilon y. \quad \text{(C1)}$$

Over one step this is a linear map of (x, y) plus an offset. For $u = +1$ the map has the form $\left(\begin{smallmatrix} a & -b \\ b & d \end{smallmatrix}\right)$ and the offset is (f, g). For $u = -1$ the signs of b and g flip. The input-averaged step is therefore diagonal, the stationary mean lies on the X axis at $m = f/(1-a)$, and each input writes $u\, v$, with $v = bm + g$, along the measured axis Y.

How the orders arise. The averaged step keeps the two axes apart, while the input-dependent part swaps them. A signal written into Y must pass through X and back to pick up more inputs, so Y carries only odd orders. By the recursion (A3), covariance passes from Y to X with the factor $b^2/(1-a^2)$ and back with $b^2/(1-d^2)$. Call the product of the two factors χ. Order $2j+1$ then carries the share $(1-\chi)\chi^j$ of the capacity of Y, and

$$\mathbb{E}D = \frac{1+\chi}{1-\chi}, \qquad \operatorname{Var} Y = \frac{v^2}{(1-d^2)(1-\chi)}. \quad \text{(C2)}$$

Single past inputs carry the share $d^2(1-\chi)$.

Fixed turn. Let the relaxation slow down, $\epsilon \to 0$, at a fixed angle θ with $\sin\theta \neq 0$. Then $1 - \chi = 3\epsilon/\sin^2\theta + O(\epsilon^2)$: the order grows as $1/\epsilon$, ϵD becomes exponentially distributed with rate $3/(2\sin^2\theta)$, and the share of single inputs vanishes, even for exact expectation values.

Gentle writing. If instead θ^2/ϵ approaches a constant ν, then to first order $a = 1 - (1+\nu/2)\epsilon$ and $d = 1 - (1+\nu)\epsilon/2$, with $b \approx \theta$ and $f \approx \epsilon$, and as $\epsilon \to 0$

$$\chi \to \frac{\nu^2}{(\nu+2)(\nu+1)}, \qquad \operatorname{Var} Y \to \frac{4\nu}{(\nu+2)(3\nu+2)}. \quad \text{(C3)}$$

The mean order stays finite, and single past inputs keep the share $1-\chi$. Their recall beyond delay τ is the fraction $d^{2\tau}$ of the total, so the range that holds 90% of it grows as $1/\epsilon$: $\epsilon\tau_{90} \to \ln 10/(1+\nu)$.

The best turn. The signal of Eq. (C3) is largest at $\nu = 2/\sqrt{3}$ (Fig. 4d). For $\nu \gg 1$ it vanishes as $4/(3\nu)$, because later inputs mix earlier ones into products. For $\nu \ll 1$ it vanishes as ν, because each input writes too little. Among power laws $\theta \propto \epsilon^\alpha$, only $\alpha = 1/2$ keeps a finite signal: the turn must fall as the square root of the relaxation rate, that is, as one over the square root of the memory lifetime. Direct propagation of the density matrix reproduces all these expressions.

D Three spins at strong loss

What is certified. Take three spins with a collective channel of opposite phases, Eq. (D1), four Hamiltonians H1–H4 with cou-

plings drawn from fixed seeds, and every Hamiltonian whose three couplings lie in a small box around theirs. As γ grows:

- At a fixed input amplitude, the order of the memory of X_1, divided by γ, becomes exponentially distributed, and the signal in X_1 falls as $1/\gamma$ (boxes of half-width 1/32).
- With the amplitude scaled as $8/\sqrt{\gamma}$, single past inputs keep a finite share of the capacity of X_1, and the range of delays that holds 90% of their recall grows in proportion to γ (boxes of half-width 1/512).

Model. The three spins evolve under

$$H_\eta(u) = \sum_{i<j} J_{ij} X_i X_j + \tfrac{1}{2} \sum_i Z_i + \tfrac{3+\eta u}{4} \sum_i X_i \tag{D1}$$

with one jump $\sqrt{\gamma}\,(\sigma_1^- - \sigma_2^- - \sigma_3^-)$, steps of $\Delta t = 1/2$ and signs $u = \pm 1$, at $\eta = 1$ (fixed amplitude) or $\eta = 8/\sqrt{\gamma}$ (gentle writing). This is the field of Sec. 2 with signs in place of uniform inputs. The jump leaves three dark states: the state with no excitation and two states with one excitation. For the four Hamiltonians the certified rate of the exponential law at a fixed amplitude lies between 61 and 133. With gentle writing at $\gamma = 2^{16}$, single inputs keep between 8% and 81% of the capacity of X_1, and the range of their recall lies between 0.18γ and 0.50γ.

Idea. At strong loss, one step of the full master equation is a rotation of the three dark states, up to corrections of order $\epsilon = 1/\gamma$. The mixing law without a reset (Appendix B.5) then gives the first statement, and an expansion in six slow covariance directions gives the second. Both end in finitely many strict inequalities, which a computer checks with rigorous error bounds. The note `three_spin_limits.pdf` in the repository writes out the derivation.

Why coherence carries the memory. With gentle writing the input enters one step at order $\sqrt{\epsilon}$ and the leak at order ϵ. The leading part of the step is a rotation generated by the static Hamiltonian restricted to the dark states. It leaves only the populations of the energy levels of this Hamiltonian unchanged, and at order $\sqrt{\epsilon}$ a new input turns a population into a coherence between two levels. Dephasing in this energy basis after every step removes these coherences. Restricted to populations, a new input writes only a change of size $\epsilon^{3/2}$, the slow population dynamics sum its square over $1/\epsilon$ steps, and the stored signal is of order ϵ^2: the collapse as $1/\gamma^2$ of Sec. 3.3. Numerically, we apply the loss-only projector $\mathcal{P}$ of Appendix A.3 after every step, with or without this dephasing. The variance of X_1 stays between 10^{-4} and 2×10^{-3} from $\gamma = 2^8$ to 2^{18} when the coherences are kept and falls as γ^{-2}, by six orders of magnitude, when they are erased. At a fixed input amplitude it falls as $1/\gamma$.

What the computer checks. The archive `spin_certificates.zip` in the repository contains the generators, the exact couplings and the interval programs. They enclose every quantity in intervals with rigorous rounding (ball arithmetic of Arb, python-flint 0.9.0) and check each strict inequality of the derivation at 256 bits. A second program, sharing no routine with them, builds the full finite-rate maps and agrees with the certified rates and limiting recalls to better than one part in a million.

E Numerical details and supporting checks

This appendix backs the main text, section by section, with what we checked and what is needed to reproduce it.

E.1 Memory lifetime and dark states (Sec. 3.1)

Figures smooth recall curves for display only: each delay is averaged with its neighbours, with weights that fall linearly to zero at a fifth of the delay, and no sum uses the smoothed values. The maps of Fig. 2a,b are computed at $\gamma = 0.1$, 0.25 and 2^{-1} to 2^6 in powers of two and interpolated by shifting each recall profile along the delay. Figure 2c stops the equal-phase channel and the two collective channels at $\gamma = 64$, where their recall at the full amplitude starts to reach past the 300 delays we sum. Summed over all delays with exact moments (Appendix A.4), their capacity keeps rising up to $\gamma = 256$. With gentle writing, delays beyond 300 add about 1% to the capacity of the opposite-phase channel.

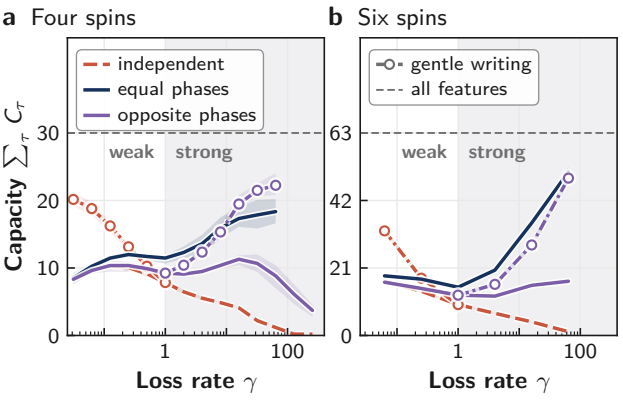


Figure 6: **Other reservoir sizes.** Capacity against the loss rate as in Fig. 2c, for four and six spins, with the same shading. A line stops where its recall reaches past the 300 delays summed.

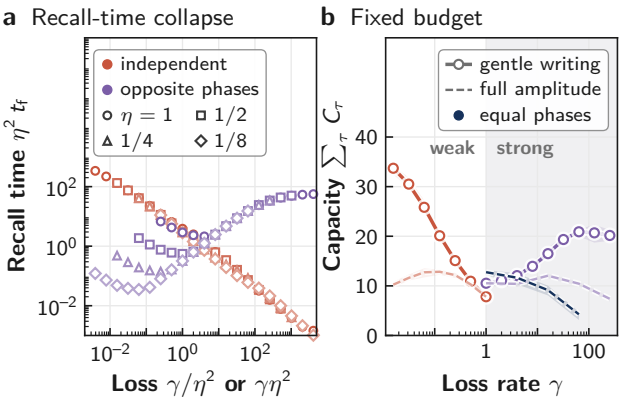


Figure 7: **Two supporting results.** **a** The time over which one input stays readable, rescaled as the sum of decay and mixing predicts: the four amplitudes fall on one curve per channel. **b** Capacity recovered at a budget of 10^9 reservoir steps per output step, with every replayed step counted. Shading as in Fig. 2c.

Both tests of the dark states act on the 100 dark coordinates. For the dark-state model we write $\mathcal{P} = RR^\flat$ with $R^\flat R = 1$, where R embeds the dark coordinates α, evolve α under Eq. (A2) and record the same 45 features of the full state $(R - \mathcal{SH}(u)R/\gamma)\,\alpha$. The readout is trained on them exactly as for the full reservoir, and the model has no fitted parameter (Fig. 3a). For the erasure test we replace the dark coordinates after every step by their time average, $r \mapsto r - RR^\flat(r - \bar{r})$. The control erases the dark subspace rotated by a random product of single-spin unitaries. Recall is measured at delays 1 to 100 with 800 training, 200 validation and 800 test inputs, at $\gamma = 16$ (Fig. 3b).

The decay out of the dark states falls exactly as $1/\gamma$: for the equal-phase channel $\gamma\bar{\kappa}$ approaches $\kappa_c = 2.945$ from $\gamma = 512$ on, so t_f grows in proportion to γ, while it levels off near 56 steps with the opposite-phase channel and near 110 with pairwise decay. The second-order calculation of Eq. (A2) gives the same κ_c at five spins and 1.4, 2.1 and 3.4 at three, four and six spins: at every size we computed, the decay falls as $1/\gamma$, with a prefactor that depends on the size.

Figure 6 repeats the curves of Fig. 2c at four and six spins, with 30 and 63 features, on eight and four reservoirs. The opposite-phase channel reverses the sign of the last $\lfloor N/2 \rfloor$ spins. With the equal-phase channel the capacity grows with the loss at both sizes, from 11.5 to 18.4 units between $\gamma = 1$ and 64 at four spins and from 15 to 51 at six, and gentle writing raises the best capacity of the other two channels. At three spins, which have only three dark states, the capacity of the equal-phase channel does not grow.

E.2 Recall time and phases (Secs. 3.2 and 3.3)

The single-input recall time t_f shows the balance of decay and mixing. Its rate adds the decay of the stored signal and a mixing rate that grows as η^2 (Appendix A.3). The decay grows as γ with independent decay and falls as $1/\gamma$ with the opposite-phase channel, so $\eta^2 t_f$ depends on loss and input only through γ/η^2 and through $\gamma\eta^2$, respectively. Both collapses hold for the four amplitudes $\eta = 1$, $\frac{1}{2}$, $\frac{1}{4}$ and $\frac{1}{8}$ (Fig. 7a), for γ from 2^{-8} to 2^6 with independent decay and from 2^{-2} to 2^{12} with the opposite-phase channel. They do not depend on how the recall time is defined: with the slowest mode in place of the mean rate they hold to within 11% wherever the recall time exceeds four steps.

The write per input of Fig. 4b is $2\|R^\flat[E(\frac{3}{4}) - E(\frac{1}{4})]r_*\|$ at the stationary state r_* of the equal-phase channel, for the input patterns $(1, \ldots, 1)$ and $\sqrt{2}\cos(2\pi i/5)$ multiplying X_i. The aligned pattern falls as $1/\gamma$, to 0.003 at $\gamma = 128$, while the other falls from 0.40 to 0.18. The uniform input of Eq. (1) is aligned with the equal-phase channel, and its single-input recall time keeps growing in proportion to γ. It is not aligned with the opposite-phase channel, and its recall time levels off. The rule is sharp: a phase error δ couples dark states directly at order δ, so beyond $\gamma \sim 1/\delta$ the equal-phase channel behaves like the opposite-phase channel. The opposite-phase

channel is the equal-phase channel with the field on spins 4 and 5 reversed, up to a relabelling of the random couplings (a turn of these spins by π about Z).

E.3 Inputs mixed and capacities (Secs. 3.2 and 3.3)

The distribution of the number of inputs mixed, Eq. (A3), uses six-point Gauss–Legendre quadrature for the expectations over the input and solves the equations $X - \bar{A}X\bar{A}^{\mathsf{T}} = C$ in the eigenbasis of $\bar{A}$. The shares of the 45 features are summed until the rest falls below 10^{-8}, or until successive orders shrink at a settled ratio, whose geometric tail we add. The linear recall follows from the same moments, $C_\tau = c_\tau^{\mathsf{T}}\Sigma_y^{-1}c_\tau/\operatorname{Var} u$ with c_τ the covariance of the features with $u_{t-\tau}$ and Σ_y their covariance, and reproduces the simulated capacities to a median of 2%. Three steps of inverse iteration on $\mathcal{T}$, from the input's covariance V and from the identity, give t_{mem}, Q and ℓ. Loss rates run in powers of four over the ranges of Fig. 4c, and with gentle writing to 16384 and 1/4096, on four reservoirs, two at the longest memory lifetimes. At the full amplitude the mean order of the opposite-phase channel rises from 1.8 at $\gamma = 1$ to 23 at $\gamma = 4096$, and with independent decay it stays within 2.2 of $s\,t_{\mathrm{mem}}$ up to $s\,t_{\mathrm{mem}} = 18$.

Figure 4a shows the linear recall summed over all delays, from the same moments. At the full amplitude it peaks at 16.0 units with the opposite-phase channel ($\gamma = 64$, $s\,t_{\mathrm{mem}} = 0.32$) and at 14.9 with independent decay ($\gamma = 1/16,\ 0.20$), and falls to 2.2 and 3.7 units at lifetimes of 750 and 430 steps, where single past inputs carry 0.05 and 0.09 of the capacity. With gentle writing single inputs carry 0.82 of the capacity at a lifetime of 20 steps and 0.87 to 0.90 beyond 80 steps, the recall stays between 39 and 40 units, and $s\,t_{\mathrm{mem}}$ between 0.007 and 0.014.

Table 3 collects the capacities. With exact readout all capacities of a readout add up to its rank, 45 here [29], so raising the linear memory capacity from 15 to 35 units leaves at most 10 instead of 30 for products of inputs.

E.4 Best input amplitude (Secs. 3.3 and 3.4)

We mapped the linear memory capacity over loss rates and amplitudes: the first four reservoirs, independent decay at $\gamma = 2^{-8}$ to 2^{-2} and at 1, the opposite-phase channel at $\gamma = 1$, 4 and 2^3 to 2^8, and fifteen amplitudes from $\eta = 2^{-7}$ to 1, with exact features or 10^6 to 10^8 preparations per setting and step from the state with the full history. With exact readout the capacity rises as the input weakens, up to 43 of the 45 units. With 10^6 to 10^8 preparations an optimum exists. Its amplitude falls with the decay-only lifetime, the single-input recall time at $\eta = 2^{-8}$, with fitted exponents between -0.40 and -0.46 in all six fits over the loss rates whose lifetime is at least four steps. Without the loss rates whose recall reaches past the 300 delays summed, the exponents are -0.48 to -0.56 with independent decay and -0.38 to -0.39 with the opposite-phase channel. The optimum is flat: one amplitude for all loss rates recovers on average 91% to 97% of the capacity at the optimum tracked at each loss rate, so the input need not be tuned to the loss rate. With independent decay the largest capacity of the map is 26.6, 32.7 and 37.2 units at 10^6, 10^7 and 10^8 preparations, at $\gamma = 1/128$ to $1/64$ inside the sampled range. With the opposite-phase channel it levels off at 15.8, 20.3 and 24.3 units from $\gamma = 64$ on.

Table 3: Linear memory capacity $\sum_\tau C_\tau$ of the five-spin reservoir in units (ceiling 45): with exact readout, at a budget of 10^9 reservoir steps per output step, and with n preparations per setting and step of replayed inputs (Sec. 3.4). Mean and standard deviation over eight reservoirs.

	independent decay	opposite phases	equal phases
tuned loss, $\eta = 1$[a]	14.5 ± 1.0	14.7 ± 1.1	30.6 ± 1.4
gentle writing[b]	34.9 ± 2.0	35.7 ± 3.0	–[c]
budget 10^9, $\eta = 1$[d]	12.8 ± 0.7	12.0 ± 0.7	12.8 ± 0.7
budget 10^9, gentle[e]	33.7 ± 1.1	20.9 ± 1.1	–[c]
$\gamma = 64$, exact[f]	1.2 ± 0.0	14.7 ± 1.1	30.6 ± 1.4
$\gamma = 64$, $n = 10^4$[f]	0.2 ± 0.0	8.2 ± 0.8	0.2 ± 0.0
$\gamma = 64$, $n = 10^6$[f]	0.2 ± 0.0	9.9 ± 0.7	2.3 ± 1.0

[a] At $\gamma = 0.1$, 64 and 64. [b] At $\gamma = 1/64$ and 256. [c] The uniform input already writes gently in this channel. [d] At $\gamma = 1/8$, 16 and 1, the best loss rates on the grid. [e] At $\gamma = 1/64$ and 64. [f] At $\eta = 1$.

E.5 Measurement (Sec. 3.4)

Weak signals cost measurements: with independent decay at $\gamma = 64$ the previous input moves the state by about 10^{-9}, which gives perfect recall of that input with exact readout and none with 10^8 preparations per setting and step. With the

Table 4: The eight decay channels at a decay-only lifetime of 32 input steps and $\eta = 0.35$: loss rate, write strength, and linear memory capacity with exact readout and with 10^6 preparations per measurement setting and step. Mean and standard deviation over eight reservoirs.

channel	loss rate γ	write strength	capacity exact	capacity at 10^6
independent	0.092	1.3	23.7 ± 1.1	17.4 ± 2.3
exchange	0.064	0.30	25.1 ± 0.8	10.1 ± 0.9
gain/loss	0.079	0.46	24.4 ± 0.8	12.5 ± 0.9
pairwise	168	1.5	18.2 ± 2.0	12.4 ± 1.2
opposite phases	101	0.68	25.3 ± 2.6	15.0 ± 1.5
two channels	73	0.011	26.6 ± 2.9	0.3 ± 0.1
equal phases	47	0.018	35.3 ± 0.8	0.3 ± 0.1
dephasing	143	0	0.0	0.2[a]

[a] A readout without any recall reads about 0.2 units, the floor of the sum over 300 delays.

opposite-phase channel, gentle writing lowers the signal from 8×10^{-5} to 5×10^{-6}, and restarting each run from $|00000\rangle$, as the replay does, raises it to 4×10^{-5}, against 1.2×10^{-4} with independent decay and gentle writing at $\gamma = 1/32$. At the full amplitude the signal of the equal-phase channel falls from 2×10^{-4} at $\gamma = 1$ to 4×10^{-7} at $\gamma = 64$.

The replayed range τ_{90} grows with the memory lifetime: along the gentle line of the opposite-phase channel from about 30 steps at $\gamma = 16$ to 130 at $\gamma = 256$. The thresholds of Sec. 3.4 are the crossings of the capacities at the full amplitude and with gentle writing, from replayed runs at 10^3 to $10^{8.5}$ preparations per setting and step. With replayed inputs the equal-phase channel at $\gamma = 64$ stays at the floor up to 10^4 preparations and recovers 2.3 units at 10^6 and 15.5 at 3×10^8. There it passes the opposite-phase channel at the full amplitude, 11.0 units, and stays below it with gentle writing, 26.3 units. Independent decay at $\gamma = 64$ stays at the floor for every number of preparations.

Figure 7b shows the capacity at a budget of 10^9 reservoir steps per output step, which is 3×10^6 to 3×10^7 preparations per setting and step, and Table 3 lists its best values. With the full history, eight times more training inputs change the sampled capacity by less than 1.3 units.

For the matched decay-only lifetime of Fig. 5b,c and Table 4 we find, for every channel and reservoir, the loss rate at which $t_{\text{leak}} = 32$ steps, at weak loss without protected operators and at strong loss with them, and run the reservoir there with $\eta = \sqrt{4/32}$. The write strength is the Euclidean norm of the Pauli coefficients $\mathrm{Tr}(PW)$, over all Pauli strings P, of the change W: the commutator $-i[H_{\text{in}}, \rho_*]$ with $H_{\text{in}} = h \sum_i X_i$ and ρ_* the stationary state, projected spectrally onto the slow modes. These are the protected ones, or without protected operators those decaying at most four times faster than the slowest. The loss rate is matched for each reservoir, so Table 4 gives its mean. Over the 56 reservoirs of the seven channels that write, the rank correlation of the write strength is 0.96 with the recovered fraction of the capacity and 0.83 with the recovered capacity itself.

E.6 Ordinary noise

Adding independent decay $\sqrt{\gamma_{\text{loc}}}\,\sigma_i^-$ to the equal-phase channel gives $\bar{\kappa} = \kappa_{\text{c}}/\gamma + \kappa_{\text{l}}\gamma_{\text{loc}}$, where to first order κ_{l} is the trace of the added dissipator on the 99 slow dark modes divided by 99, $140/99 \approx 1.414$. Local dephasing $\sqrt{\gamma_\phi}\,Z_i$ adds $\kappa_\phi\gamma_\phi$ with $\kappa_\phi = 403.2/99 \approx 4.073$, and the full spectra reproduce both coefficients to within 0.2%. A single-input recall time of at least T steps, $\kappa_{\text{c}}/\gamma + \kappa_{\text{l}}\gamma_{\text{loc}} \le 1/(\Delta t\, T)$, then needs $\gamma/\gamma_{\text{loc}} \ge 4\kappa_{\text{c}}\kappa_{\text{l}}\,\Delta t^2 T^2 \approx 4.2\,T^2$, or $12\,T^2$ for dephasing.

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